

Hicksian Decomposition of Substitution and Income Effects

JEB104 Microeconomics I
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Decomposition of Total Price Effect

Impact of a price change can be separated into 2 components:

- ➊ Substitution effect
- ➋ Income effect

2 methods:

- ➊ Hicksian decomposition (now)
- ➋ Slutsky decomposition (later)

Previously

- Impact of price change and what it means

Today

- Decomposition (what's behind the decision after a change?)

In the previous lecture, we analyzed what happens after a price change, but now we will analyze what is behind in a greater detail

- When the price changes, income also “changes”

Draw figure:

- Change in price, change in optimum
- Movement from one optimum to the other
- Separated effects of changes in relative prices and relative income

We will introduce Hicksian method of decomposition

- Slutsky decomposition will be done later

Varian: chapter 8 – Slutsky decomposition (Hicksian decomposition in subsection 8.8 having much less attention)

Income Effect

↓ price of x_1 ($p_1^0 \rightarrow p_1^1$) $\Rightarrow$ consumer can buy more of x_1 and also x_2 $\Rightarrow$

$\Rightarrow$ ↑ of real income: $\Delta m_{real} = m_{p^1, u^1} - m_{p^1, u^0}$

① ↓ p ($p^0 > p^1$) $\Rightarrow \Delta m_{real} > 0$

② ↑ p ($p^1 > p^0$) $\Rightarrow \Delta m_{real} < 0$

Income effect = real-income-induced change in demand

$$IE = \Delta m_{real} \cdot \frac{\Delta D_i(p, m)}{\Delta m}$$

$$IE = \Delta p_1 \cdot \underbrace{\frac{\Delta m_{real}}{\Delta p_1}}_{<0} \cdot \underbrace{\frac{\Delta D_1(p, m)}{\Delta m}}_{>0 \text{ normal, } <0 \text{ inferior}}$$

Normal good: **opposite sign** of income effect and price change

Inferior good: **same sign** of income effect and price change

Substitution Effect

↓ price of $x_1 \rightarrow$ change in relative prices (flatter budget line)

x_1 now relatively cheaper than $x_2 \rightarrow$ consumer demands more of x_1 as he substitutes x_1 for x_2

Substitution effect

- change in demand from substitution when relative prices change, but there is no change in real income (returning the consumer to the original indifference curve)
- always opposite direction than price change: ↓ price $\rightarrow$ ↑ quantity and viceversa (from convexity of preferences)

$$SE = \Delta p_i \cdot \underbrace{\frac{\Delta x_i^*}{\Delta p_i}}_{<0} \Big|_{u(x)=u^0}$$

Δm_{real}

- m_{p^1, u^1} : With how much money we can reach u^1 when prices are p^1 = it is income we have.
- m_{p^1, u^0} : How much money we would theoretically need to reach u^0 under the new prices p^1 .

Change in real income is positive for price decrease and negative for price increase

$$\frac{\Delta D_i(p, m)}{\Delta m}$$

- It is “the same” as $\frac{\partial D_i(p, m)}{\partial m}$, so it tells you how demand changes with change in income (rate)
- Slope of Engel curve
- Flat part $\rightarrow$ not large change in demand with change in real income
- Steep part $\rightarrow$ high change in demand with given change in real income

$$\frac{\Delta m_{real}}{\Delta p_1} < 0 \text{ real income and price change always in the opposite direction (see above)}$$

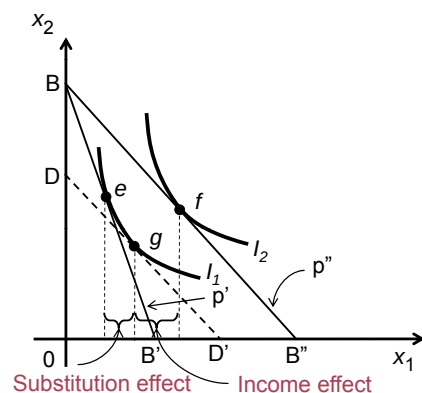
Change in demand induced by income effect has an opposite sign than price change for normal goods and the same sign as price change for inferior goods

Why change in demand from SE moves in the opposite direction than price change

- Assume $p_1 \downarrow$
- **Before the price change:** consumer must give up A units of x_2 to obtain 1 unit of x_1 in the market, given market prices (follows from a slope of the budget line, hence $-\frac{p_1^0}{p_2} = -A$)
- Would be willing to do it in this way to keep u^0 : $(x_1 + 1, x_2 - A) \sim (x_1, x_2)$
- In the original optimum (x_1^0, x_2^0) : $|MRS^0| = \left| -\frac{p_1^0}{p_2} \right| = A$
- **After the price change** (keeping u^0): x_1 is cheaper now, so consumer has to give up $A - \epsilon$ units of x_2 to obtain 1 unit of x_1 in the market, given new market prices ($\epsilon > 0$)
- In the original optimum (x_1^0, x_2^0) : $|MRS^0| = A > \left| -\frac{p_1^1}{p_2} \right| = A - \epsilon \rightarrow \frac{MU_1^0}{MU_2^0} > \frac{p_1^1}{p_2}$
- In the new optimum (x_1^1, x_2^1) : $|MRS^1| = \left| -\frac{p_1^1}{p_2} \right|$
- Hence, $|MRS^1| = \frac{MU_1^1}{MU_2^1} < \frac{MU_1^0}{MU_2^0} = |MRS^0|$ and $x_1^1 > x_1^0$, $x_2^1 < x_2^0$ (using $MU \downarrow$ with $\uparrow x$ which follows from convexity of preferences)
- So, with $\downarrow p \uparrow x$ and with $\uparrow p \downarrow x$ when utility is kept constant – follows from the shape of indifference curves (convex preferences = diminishing MRS)

If it is not the case, preferences cannot be convex!

Income and Substitution Effects



- (i) The *substitution effect* is measured by movement along the indifference curve from *e* to *g*.
- (ii) The *income effect* of the price change is measured by the parallel shift of the budget line from *DD'* (point *g*) to *BB''* (point *f*).

Income and Substitution Effects

Total price effect:

$$\overbrace{\Delta p_1 \cdot \frac{\Delta D_1(p, m)}{\Delta p_1}}^{TE} = \underbrace{\Delta p_1 \cdot \frac{\Delta x_1^*}{\Delta p_1} \Big|_{u(x)=u^0}}_{<0}^{SE} + \underbrace{\Delta p_1 \cdot \frac{\Delta m_{real}}{\Delta p_1} \cdot \frac{\Delta D_1(p, m)}{\Delta m}}_{<0}^{IE}$$

$$\frac{\Delta D_1(p, m)}{\Delta p_1} = \underbrace{\frac{\Delta x_1^*}{\Delta p_1} \Big|_{u(x)=u^0}}_{<0}^{SE} + \underbrace{\frac{\Delta m_{real}}{\Delta p_1} \cdot \frac{\Delta D_1(p, m)}{\Delta m}}_{<0}^{IE}$$

Good	TE	SE	IE
Ordinary, normal	–	–	–
Ordinary, inferior	–	–	+
Giffen, inferior	+	–	+

Giffen good is always inferior!!!

DRAW

1. Standard choice: Old optimum – point *e*
2. $p_1 \downarrow$: New optimum – point *f*
3. Draw imaginary BC with new prices that touches the old IC: Intermediate optimum – point *g*
4. IE: intermediate – new
5. SE: old – intermediate

You might watch this [video](#).

Formulas are just rewritten from the previous slides

Total price effect

- change in demand due to change in price (movement from the point *e* to the point *f* in the graph above)
- can be decomposed into substitution and income effect

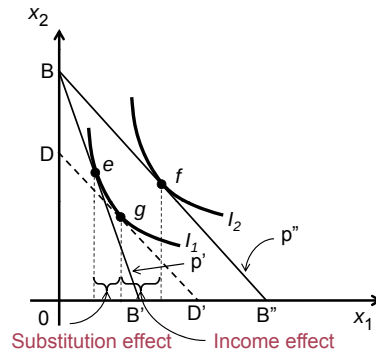
We can then eliminate Δp_1 (it is everywhere) – so instead of absolute changes we look at rates how demand changes (2nd formula)

Left-hand side of the 2nd formula represents the slope of Walrasian demand! It is negative for ordinary good (downward-sloping demand) and positive for Giffen good (upward sloping demand).

Giffen good is always inferior!!!

- Because the substitution effect is always negative, the income effect has to be stronger than the substitution effect in order to “push” the total effect to be positive
- income effect is strong only for goods which comprise a large part of a budget
- the good can be Giffen only for specific price and income levels (it can be an ordinary good for other price-income combinations)

Ordinary and Normal Good



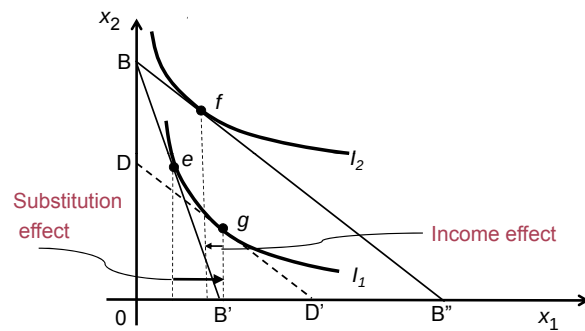
SE ($e \rightarrow g$) and IE ($g \rightarrow f$) work in the same direction
 $\downarrow$ of price $\Rightarrow$ SE: $\uparrow$ demand, IE: $\uparrow$ demand

The same picture as above.

Ordinary and normal good

- Both income and substitution effects work in the same direction
- With decrease of price of x_1 , the amount of x_1 demanded increases due to SE and also IE.

Ordinary and Inferior Good

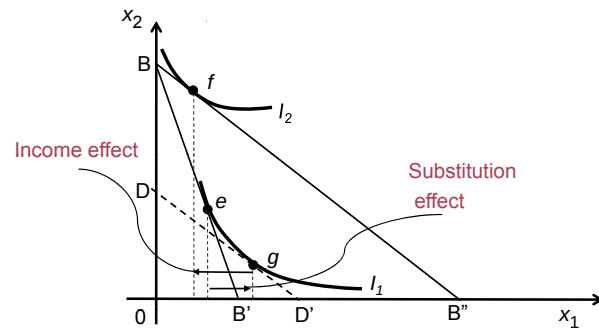


SE ($e \rightarrow g$) and IE ($g \rightarrow f$) work in the opposite direction
 $\downarrow$ of price $\Rightarrow$ SE: $\uparrow$ demand, IE: $\downarrow$ demand, $|SE| > |IE|$

Ordinary and inferior good

- SE: $p_1 \downarrow \Rightarrow x_1 \uparrow$ (move from e to g)
- IE: good 1 is an inferior good: $m_{real} \uparrow \Rightarrow x_1 \downarrow$
- Therefore, the inferiority of the good partly offsets the substitution effect (g to f)

Giffen Good



SE ($e \rightarrow g$) and IE ($g \rightarrow f$) work in the opposite direction

$\downarrow$ of price $\Rightarrow$ SE: $\uparrow$ demand, IE: $\downarrow$ demand, $|IE| > |SE|$

Cross-Price Effects

Change in price of $x_i \rightarrow$ change in demand for x_j

$$\overbrace{\Delta p_i \cdot \frac{\Delta D_j(p, m)}{\Delta p_i}}^{CTE} = \overbrace{\Delta p_i \cdot \frac{\Delta x_j^*}{\Delta p_i} \Big|_{u(x)=u^0}}^{CSE} + \underbrace{\Delta p_i \cdot \frac{\Delta m_{real}}{\Delta p_i}}_{<0} \cdot \frac{\Delta D_j(p, m)}{\Delta m}$$

Net substitutes: $\frac{\Delta x_j^*}{\Delta p_i} \Big|_{u(x)=u^0} > 0$

Net complements: $\frac{\Delta x_j^*}{\Delta p_i} \Big|_{u(x)=u^0} < 0$

① x_j net complement of x_i and normal $\Rightarrow x_j$ gross complement of x_i

② x_j net substitute of x_i and inferior $\Rightarrow x_j$ gross substitute of x_i

If $|CSE| > |IE|$:

③ x_j net complement of $x_i \Rightarrow x_j$ always gross complement of x_i

④ x_j net substitute of $x_i \Rightarrow x_j$ always gross substitute of x_i

Giffen

- The income effect is so strong that it suppresses the substitution effect (remember that the good is INFERIOR!)
- income effect becomes stronger as we spend more on the good

Real-life example:

- Rice purchased by relatively poor people in Asian countries.
- Poor people eat mainly rice, spend large part of their income on rice
- If its price decreases, their relative wealth increases and they can afford to buy some meat.
- income effect causes that they increase their consumption of meat and decrease their consumption on rice.
- Paper by Jensen and Miller (2008) in American Economic Review.

The changes work in a similar way, but now it's cross price change

LHS: Total cross-price effect - so how Walrasian demand for x_j changes with price of x_i . It is positive if goods are gross substitutes and negative if goods are gross complements. (remember that x_i might be gross substitute of x_j while x_j is gross complement of x_i , so it is not mutual relation)

RHS: Cross-price substitution effect - a mutual relation - if x_i net complement of x_j , then x_j net complement of x_i

Net complements or substitutes in the case of 2 goods?

- Original optimum x_1^0, x_2^0 : $MRS^0 = -\frac{p_1^0}{p_2^0}$
- $\uparrow p_1$, $p_1^1 > p_1^0$: $|MRS^0| < |-\frac{p_1^1}{p_2^0}|$, hence new optimum while keeping u^0 : x_1^1, x_2^1 , when $x_1^1 < x_1^0$, $x_2^1 > x_2^0$
- $x_2 \uparrow$ with $\uparrow p_1 \rightarrow x_2$ is a net substitute of x_1
- similar reasoning holds for $\uparrow p_2 \rightarrow x_1$ is a net substitute of x_2
- we cannot have net complements in the case of 2 goods

All goods cannot be net complements.

- x_1 and x_2 net complements, hence $\uparrow p_1 \Rightarrow \downarrow x_2$ (cross-substitution effect), $\downarrow x_1$ (substitution effect)
- to keep utility constant, there must be x_3 increasing with $\uparrow p_1$

If, however, $|IE| > |CSE|$:

- x_j net complement of x_i and inferior: $\Rightarrow x_j$ gross substitute of x_i
- x_j net substitute of x_i and normal $\Rightarrow x_j$ gross complement of x_i

Income and Substitution Effects: Examples

Perfect complements:

- $SE=0$

Perfect substitutes:

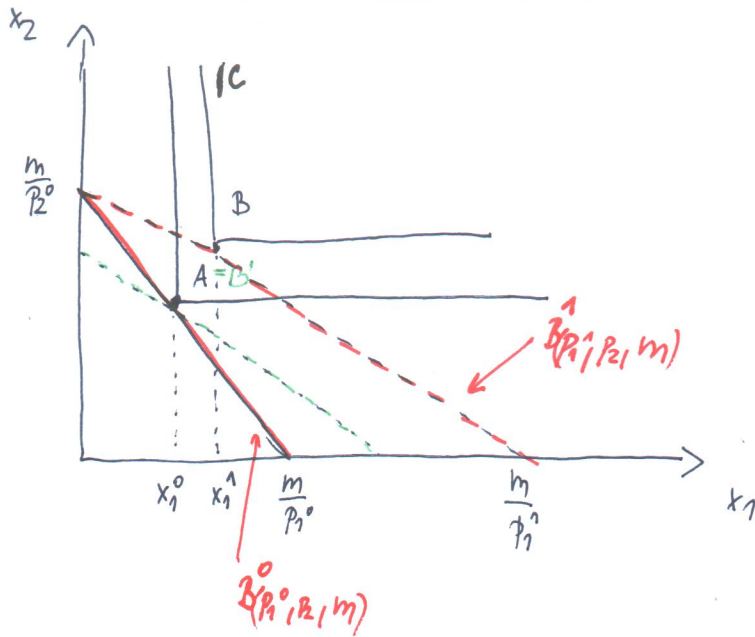
- $SE=0$ or $IE=0$

Quasi-linear preferences:

- $IE=0$

HICKSIAN SEPARATION OF SE AND IE

Example: Perfect complements



1. Original : point A

2. $p_1^0 \downarrow p_1^1 \Rightarrow$ new optimum point B

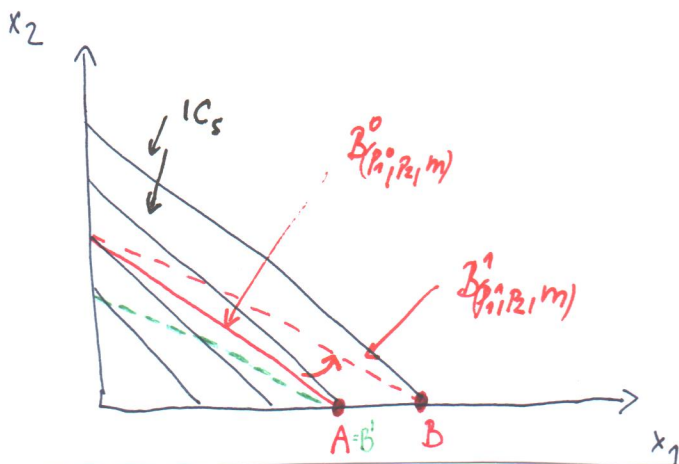
How can we decompose Δx_1 due to this price change to SE and IE?

$\Rightarrow$ no substitution effect
 $\rightarrow SE = 0 \quad (A \rightarrow B')$

$\Rightarrow$ the whole shift is due to income effect

$$IE = x_1^1 - x_1^0$$

Perfect substitutes



1. Original : point A

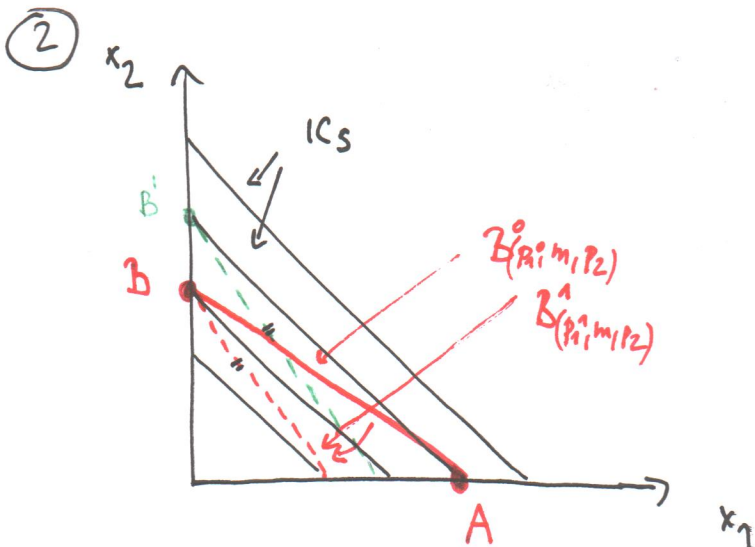
$$x_1^0 = \frac{m}{p_1^0}$$

2. $p_1^0 \downarrow \Rightarrow$ new optimum B

$$x_1^1 = \frac{m}{p_1^1}$$

$$SE = 0 \quad (A \rightarrow B')$$

$$IE = x_1^1 - x_1^0$$



1. original optimum: A

$$x_1^0 = \frac{m}{p_1^0}$$

2. $\uparrow p_1 \Rightarrow p_1^1$

Price of x_1 too high that x_1 no more consumed $\Rightarrow x_1^1 = 0$

$\rightarrow$ new optimum B

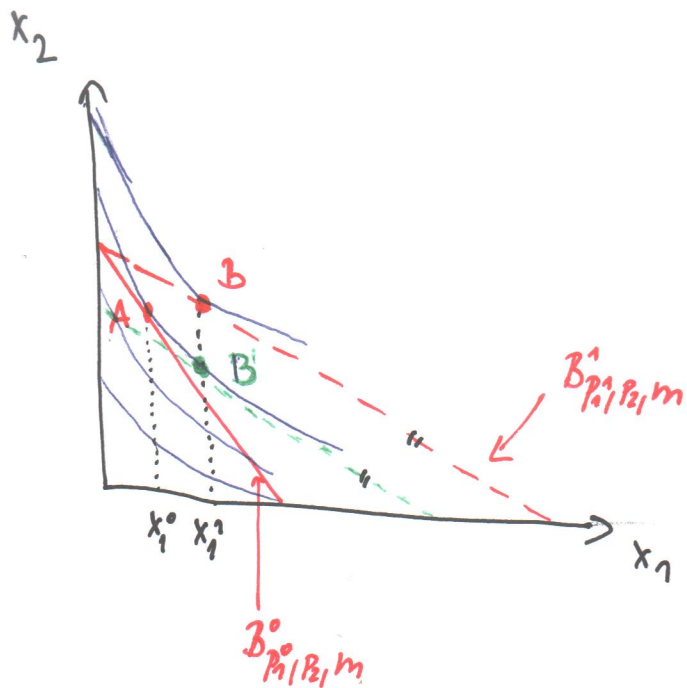
SE: $A \rightarrow B'$

$$SE = x_1^1 - x_1^0$$

$$\downarrow$$

$$IE = 0$$

Quasilinear preferences : $u(x_1, x_2) = f(x_1) + x_2$



1. Original optimum: A
 x_1^0

2. $\downarrow p_1 \Rightarrow p_1^0 \rightarrow p_1^1$

SE: $A \rightarrow B'$, $SE = x_1^1 - x_1^0$

IE: $B' \rightarrow B$

$\Rightarrow IE = 0$

Points B, B' : MRS is the same

m